



3Doodler Kit & pens

New Star Trek and Powerpuff Girls-themed pens and also a new robotics kit on the way. <http://dgit.in/3doodlerKit>



Rare Carat's new chatbot

Price transparency website for consumers buying jewellery adds new Watson-powered chatbot. <http://dgit.in/CaratWatson>

From the labs

ion system reaches its ground state – a state of equilibrium. Intuitively, one would imagine that this system doesn't show any motion, similar to how the block and spring system wouldn't in the absence of disturbance. Although, if we throw in time symmetry breakdown into the mix something weird can happen. The ions can start rotating in the ring in a perpetual circling motion. The breakdown of spatial symmetry manifested into a repeated pattern in space for crystals. Similarly, a time symmetry breakdown would create periodicity of motion in time. While this might sound



Nobel Laureate Frank Wilczek who first came up with the idea of time crystals

like a perpetual motion machine, it really isn't. In this case, the ions do move around in a loop, perpetually, but this movement doesn't extract any energy from the system even in the ground state. The formidable law of conservation of energy also states that since there is no usage of energy there is also no energy available to extract for work. This ring-shaped crystal arrangement was one of the first feasible experiments postulated to create time crystals and was published along with Wilczek's original paper in 2012. The method works in theory, but there are some practical difficulties.

For one, quantum systems in equilibrium show no dependency on time and thus it is impossible for them to show any periodicity as well. To get around this, it was suggested that time crystals be created in a 'non-equilibrium' state, where a system is constantly driven by some periodic source. To keep the system from absorbing energy from this source, the constituent ions in the system had to be controlled such that they remain localised and not

smear out as the laws of probability for quantum particles dictate. A recipe based on the above considerations was suggested last year and two teams independently followed it to create a time crystal in the lab.

One of the teams from the University of Maryland used an array of ten ions, electrons with spins. Then they bombarded the system with two laser pulses timed with precision. One pulse created a magnetic field and the other flipped the electron spins. Because the system of electrons is interacting and spins are in a quantum entanglement (to put it simply: a change in one spin affects the other), this system settles into a state where the spins change periodically in space. This results into the spontaneous breaking of the spatial symmetry. As a simple picture, imagine a row with ten seats. The person on the first seat faces the movie screen, and the adjacent one faces away. This arrangement has the sort of repeating quality common to a crystal in ground state. Now after a time period, this arrangement switches and each individual changes the direction he is facing. After another interval, the system reaches back to its first state again. This signifies breakdown of time symmetry. Not only does the ion system behave like a crystal in space, but also oscillates back and forth in arrangements, in time. What is even more surprising is that if the period of the laser pulse driving is T , the arrangement of spins in the ions oscillates in time with a period of $2T$ – a behaviour that is impossible to occur in a normal system. Furthermore, on changing parameters

of the driving pulse, the system remains in time oscillations with period $2T$. Since there is no driving force with period $2T$, the only way the ions could oscillate with that period is by breaking time symmetry. Voila, a time crystal!



TECHNICALITY IS TOO DAMN HIGH

This might sound like technical gibberish but it's a

huge experimental achievement. What scientists have demonstrated is the existence of a new phase of matter, one that is never in equilibrium. A ground state of energy, carrying out a perpetual oscillation was believed to be impossible. How can a system move back and forth without using up energy? Until now, no one had broken the old guard of time symmetry. Now that it has been done, it opens the doors to some new, radical research. For instance, devices operating at the quantum level have a life time limit due to a phenomenon known as 'decoherence'. This phenomenon is responsible for bringing a quantum system back to a classical state – where it is ruled by the laws of classical mechanics. For quantum computing devices, this is an undesirable effect. Time crystals can potentially lead to devices that are stable against such effects, enabling better memory and computing devices for quantum computing applications. At a more fundamental level, since time crystals are an example of never seen before time symmetry breakdown, an experimental realisation is likely to provide scientists with access to states of matter that could only be theoretically possible in such systems. The fact that time crystals present a time based analogue to ordinary crystals that exhibit spatial symmetry breaking, indicates that maybe time as a concept must be unified with the fabric of space, something that current quantum mechanics doesn't feature. In conclusion, we are at the cusp of some great discoveries which in the long run can alter our understanding of not only the standard states of matter, but the fundamental concept of time itself.

